Week 2 of 6

2 Wallets & on-chain basics

Task	Get devnet SOL and USDC, send USDC to a partner, read the transaction on Orb.
Blog post	How I interacted with my first Solana dApp.

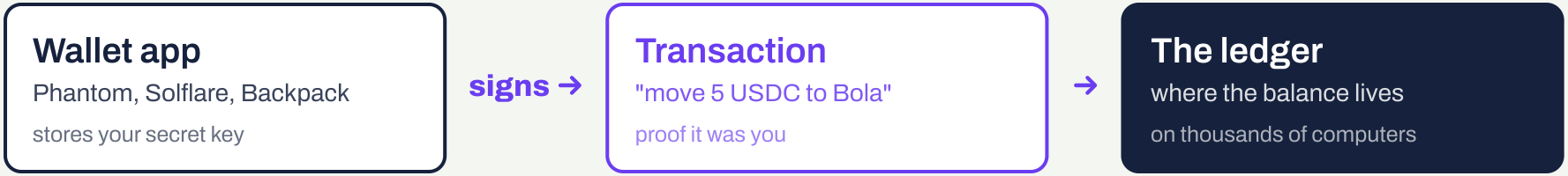
Today

1. Keys
2. Accounts
3. Tokens
4. Transactions
5. → Send devnet USDC

Keys

Your wallet doesn’t hold coins.

A wallet is a keyring, not a purse



Lose the phone, keep the phrase: your money is fine.
Lose the phrase: nobody can help you.

Twelve words, every address



Same 12 words, path `m/44'/501'/1'/0'` → a second address. One phrase can restore every account.

Sign something

A real keypair, made in this browser. Sign, then edit the message.

Generate a keypair

Public key = your address — share freely

not yet

Secret key — never share

Message

Pay Bola 5 USDC

Sign with secret key

Signature

not yet

Who holds the key?

Custodial

An exchange holds it for you.

Easy recovery. They can freeze you. If they fail, you queue with everyone else (FTX).

Non-custodial

You hold it. Phantom, Solflare, Backpack.

Full control. No password reset. Hardware wallets keep the key off your laptop.

> **The one rule**
> Nobody legitimate will ever ask for your recovery phrase. Not support, not a mentor, not a "wallet validator" site.

'Phantom Support' DMs you: there's a sync bug, enter your recovery phrase at their link to fix it. What do you do?

- **A** Enter it, support needs it to fix things
- **B** Send only the first 6 words
- **C** Ignore and block. It's a theft attempt
- **D** Move funds first, then enter it

Accounts

Everything on Solana is one.

One shape for everything

Your wallet | A token mint | Your token account | A program

address `a public key, like 7xKp…Qm2a`		
`lamports`	**2,500,000,000 (2.5 SOL)**	SOL balance, in billionths
`data`	**empty**	bytes the owner program reads
`owner`	**System Program**	the only program allowed to change it
`executable`	**false**	is this account code?

Just SOL. No data. Owned by the System Program, which only lets your key move it.

Rent is a deposit, not a fee

Create an account

Lock a small SOL deposit for the space it uses.

`token account: 0.00203928 SOL`

Use it

The deposit sits there. Nothing is charged over time.

Close it

The deposit comes back to you.

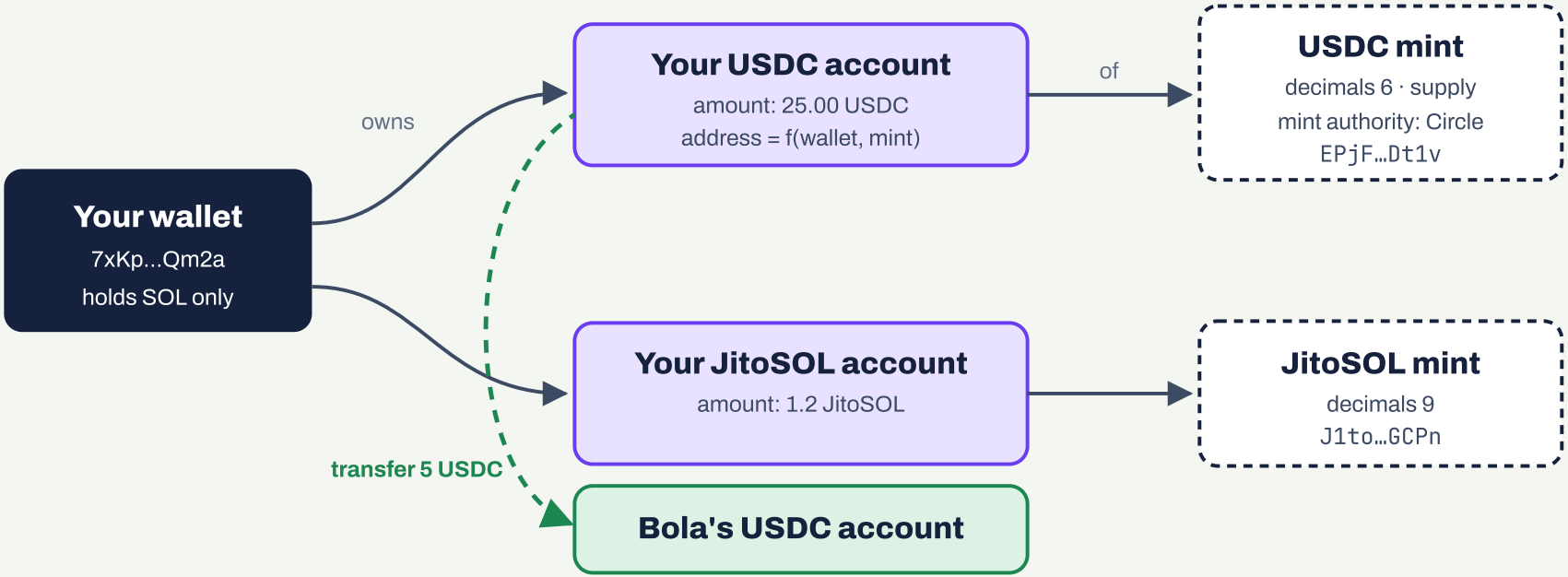
`+0.00203928 SOL`

That's why your **first** USDC send to someone costs more than the second.

Tokens

One program, many mints.

Wallet → token account → mint



Which one is USDC?



Same name, same logo. Which one is Circle's devnet USDC?

Two token programs

SPL Token

The original. Mint, transfer, burn, freeze.

USDC, most older tokens.

Token-2022

Same basics, plus optional **extensions**.

PYUSD, USDG, many newer tokens.

Transfer fee | Transfer hook | Permanent delegate | Non-transferable | Confidential transfer | Metadata

> Both programs have a **mint authority** and a **freeze authority**. Check who holds them before you buy anything.

Where does your USDC balance actually live?

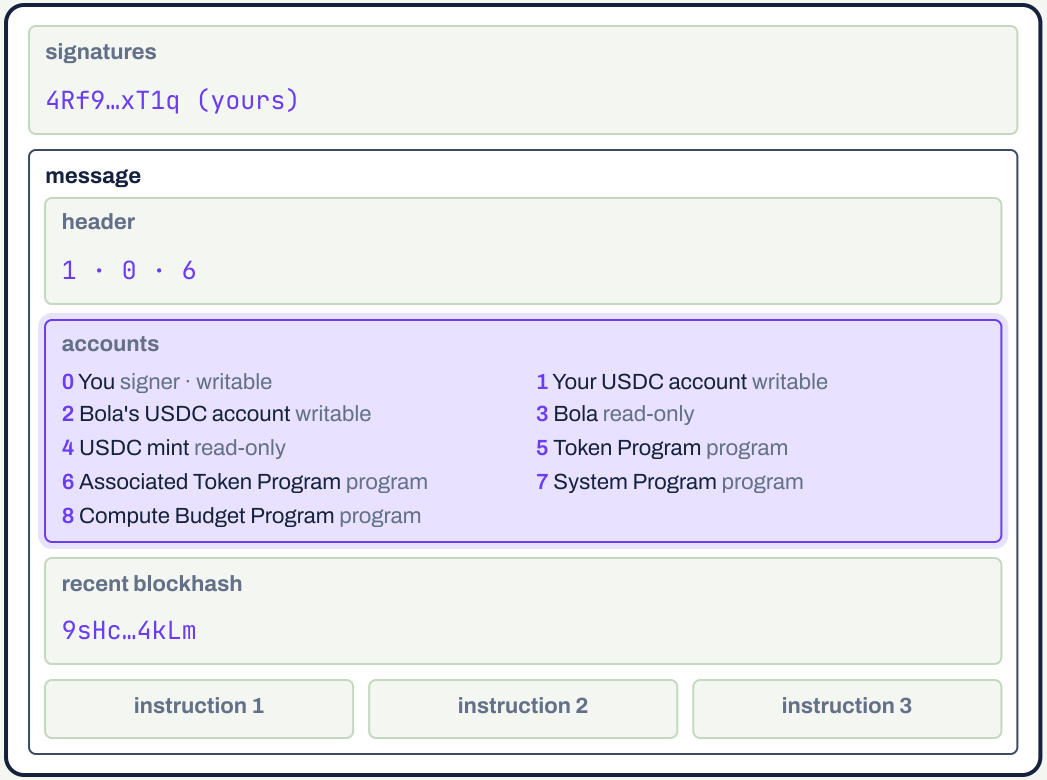
- **A** Inside your wallet account
- **B** Inside the USDC mint account
- **C** In your own USDC token account
- **D** In the Token Program's code

Transactions

What you actually sign.

Inside a transaction

Sending 5 USDC to Bola, who has never held USDC. Hover each part.



Account list

Every account the transaction will touch, listed up front. This is what lets Solana run transactions in parallel.

The life of a transaction



Build
The app lists the instructions and accounts.

Sign
Your wallet shows what it will do. You approve.

Send
Sent to an RPC node, then forwarded straight to the next leaders. No mempool.

Execute
The leader runs it in a ~400ms slot. All instructions succeed, or none do.

Processed
In a block on one node. Could still be dropped.

Confirmed
Validators holding 2/3 of stake voted for the block.

Finalized
31+ blocks built on top. Treat it as permanent.

What it costs

Signatures 1

Compute limit 200,000 CU

Priority price 10,000 μ-lamports/CU

SOL price (example) $150

☑ Creates the recipient's token account

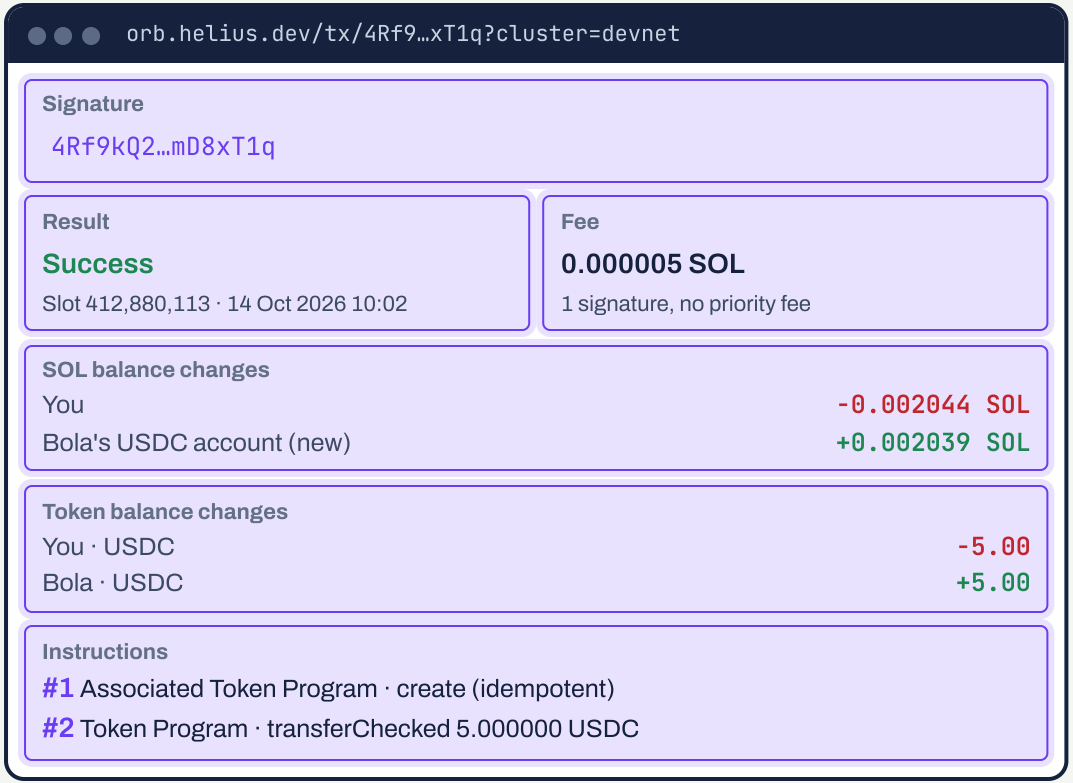
Base fee	0.000005 SOL
Priority fee	0.000002 SOL
Rent deposit	0.00203928 SOL

Fee (gone)	Deposit (refundable)
$0.00105	$0.306

Your swap fails because the price moved too much. What happens to the fee?

- **A** Refunded automatically
- **B** Charged anyway
- **C** Charged only if you retry
- **D** Paid by the DEX

Reading it on an explorer



Everything an analyst needs is on one page. Write down what you can and can't interpret.

Sample data for teaching.

This week’s task

Your first devnet transaction.

The task

Users

1. Install **Phantom** (or Solflare, Backpack). Turn on **Testnet Mode** and pick **Solana Devnet**. Check the banner.
2. Get devnet SOL: **faucet.solana.com**
3. Get devnet USDC: **faucet.circle.com**
4. Send USDC to a partner.
5. Open it on **Orb** with `?cluster=devnet`. Find: the **fee**, the **token instruction**, the **token account it created**.

Builders

Everything on the left, plus:

1. Create your own token with the `spl-token` CLI.
2. Mint 1,000 to yourself.
3. **Revoke the mint authority.**
4. Look at the mint account again. What changed?

Builders: your own token

```
solana config set --url devnet
solana-keygen new -o ~/devnet.json && solana airdrop 2

spl-token create-token
# → prints your new MINT address

spl-token create-account <MINT>
# → creates your token account (ATA) for it

spl-token mint <MINT> 1000
spl-token supply <MINT>

spl-token authorize <MINT> mint --disable
spl-token display <MINT>   # mint authority: (not set)
```

Blog post

"How I interacted with my first Solana dApp."

Include **one transaction link** and an **annotated explorer screenshot**.

Teacher prep

For the mentor. Skip in class.

You, before class

- Make a **fresh wallet** and run the whole task on it. Note which faucet rate-limits you.
- **Pre-fund a teacher wallet** with devnet SOL and USDC. Faucets break when a whole room shares one Wi-Fi.
- Prepare one whiteboard diagram: **wallet → token account (ATA) → mint**. Weeks 3 and 4 build on it.
- Read Solana’s intro to transactions in the official docs.

> **Faucet limit**
> Circle's faucet gives 20 USDC every 2 hours per address (confirmed in Circle's FAQ, 14 Oct 2026). Plan the transfer task around that.

Words from week 2

Keypair	A secret key and the public key made from it.
Recovery phrase	12 or 24 words that rebuild every key.
Account	lamports, data, owner, executable. Everything is one.
Rent	A refundable deposit for an account's storage.
Token account (ATA)	Holds one wallet's balance of one token.
Blockhash	Recent block ID that makes a transaction expire.

Address	A public key written in base58.
Signature	Proof that a key holder approved this exact message.
Lamport	One billionth of a SOL.
Mint	The account that defines a token.
Instruction	One call to one program inside a transaction.
Devnet	Solana's free test network for apps.